Ximena Aragón Toledo

FOOD ENGINEER

Curriculum Vitae

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PERSONAL DETAILS

Profession: Food Engineer
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Languages: Spanish (native) and English (C1)



PROFESSIONAL OBJECTIVE

Food Engineer from the Universidad Tecnológica de la Mixteca (Mexico) with experience in production, product development and process optimization in the food industry. I have run craft beer production, researched and developed fermented products, and carried out laboratory projects on functional beverages and on clarifying kombucha through microfiltration; I also developed the «Zumbido» mead end to end. I am driven by continuous improvement, problem solving and meeting operational goals, with a strong focus on quality assurance and food safety. I am looking to join a production, quality or new product development team where I can contribute and keep growing professionally.

AREAS OF INTEREST

- Production and process operation in the food and beverage industry.
- New product research and development.
- Quality assurance and food safety.
- Fermented and functional beverages: beer, kombucha and mead.
- Process standardization and optimization; continuous improvement.
- Sensory evaluation and consumer acceptance.

EDUCATION

Universidad Tecnológica de la Mixteca — Huajuapan de León, Oaxaca, Mexico
B.Eng. in Food Engineering — *2021 – 2026*

- Hands-on training in the Food Processing Pilot Plant and the Bioprocess Laboratory: made more than 20 products across the meat, dairy, cereal, and fruit and vegetable lines (see Projects).

PROFESSIONAL EXPERIENCE

EtnoFood — Professional internship
New Product Research & Development — *July 2025 – August 2025*

- Developed fermented products —such as kombucha— and new products such as a traditional salsa and tofu, from formulation to finished product.
- Monitored fermentation parameters and followed each batch through the process to keep it stable.
- Solved problems and optimized formulations through sensory evaluation.
- Standardized processes to ensure reproducibility and operational efficiency.
- Packaged products under quality and food-safety criteria.
- Ran sensory evaluations with panelists and analyzed consumer acceptance.

La Fermentería — Professional internship

Brewery Production *July 2024 – September 2024*

- Ran the craft beer production process: mashing, recirculation and sparging, boiling and fermentation.
- Monitored critical parameters and operated equipment to keep the process running.
- Applied Good Manufacturing Practices (GMP) and followed up on quality controls during production.
- Identified and resolved deviations through sensory evaluation of the finished product.
- Promoted and sold craft beer at fairs.

Universidad Tecnológica de la Mixteca Social service
Synthesis of Metal Sensors for Food *June 2023 – November 2023*

- Took part in the initial phase of a project synthesizing chemical sensors to detect copper, iron and zinc in food.
- Ran preliminary UV-Vis and IR spectrophotometry analyses to support the evaluation of the synthesized materials.
- Applied thin-layer and column chromatography as part of the experimental analysis.

PROJECTS

Zumbido: red-fruit mead Food Processing Pilot Plant, UTM
Alcoholic beverage development *January 2026 – May 2026*

- Developed an artisanal mead end to end —an alcoholic beverage fermented from the honey of bees native to Oaxaca's Sierra Sur, made with red fruit—, from concept to a standardized production process.
- Took the product through packaging, labeling and commercial presentation under the name «Zumbido».

Kombucha microfiltration Bioprocess Laboratory, UTM
Process development *October 2025 – December 2025*

- Optimized the microbiological stabilization and clarification of kombucha through microfiltration, making the process more efficient without compromising the product's functional properties.
- Operated a microfiltration system with a peristaltic pump and pressure control, and compared the volume and clarity of the permeate with the unfiltered sample.

Functional drink with antioxidant activity Bioprocess Laboratory, UTM
Adaptogenic drink formulation *October 2024 – December 2024*

- Formulated and optimized a functional drink made with nanche-leaf extract, raspberry pulp and honey.
- Evaluated the bioactive compounds of the extracts and selected the most effective formulation with the DPPH antioxidant assay.

Pilot-plant product development Food Processing Pilot Plant, UTM
Hands-on training across four process lines *2024 – 2025*

- **Meat:** chorizo and cooked ham; physicochemical analysis of meat products.
- **Dairy:** fresh cheese, yogurt and ice cream; platform tests on milk.
- **Cereals:** baking (bread, conchas, sweet rolls and Rosca de Reyes), pizza and fresh pasta.
- **Fruits and vegetables:** ketchup, dragon fruit chamoy, jam, pickles, canned salsas and dried fruit.

SKILLS

- **Production and processes:** production process operation, process control, process standardization and optimization, continuous improvement and problem solving.
- **Quality and food safety:** Good Manufacturing Practices (GMP), quality control, quality tools, packaging and labeling.
- **Product development:** food and beverage formulation, fermentation (beer, kombucha and mead), sensory evaluation and consumer acceptance analysis.
- **Lab techniques:** UV-Vis and IR spectrophotometry, DPPH assay, thin-layer and column chromatography, microfiltration and physicochemical analysis.

- **Software:** Microsoft Excel, Microsoft Visio and SolidWorks.

LANGUAGES

Spanish	Native language	Native
English	Foreign language	C1 (advanced)

ADDITIONAL INFORMATION

- **Soft skills:** focus on continuous improvement and operational goals, problem solving, teamwork and customer-facing experience (promoting and selling products at fairs).
- **Evidence portfolio:** more than 170 photos of processes and products made in the pilot plant, the lab and industry.

APPENDIX: PHOTOGRAPHIC EVIDENCE

A selection of photos from the projects and experience described above.



«Zumbido» mead



Mead fermentation



Microfiltration setup



Unfiltered and microfiltered kombucha



Ground nanche leaf



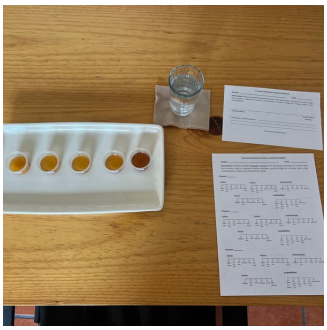
Adaptogenic drink (prototype)



Making tofu at EtnoFood



Kombucha batches at EtnoFood



Sensory evaluation



Brewhouse, La Fermentería



Fermentation tanks



Mango-strawberry jam



Fresh cheese



Chorizo



Conchas (sweet bread)



Vinegar pickles